SPhO 2025 Experiment

FIZIKA SPhO Training

November 2025

1 Problem Statement

This is recreated from memory - some small details may be inaccurate.

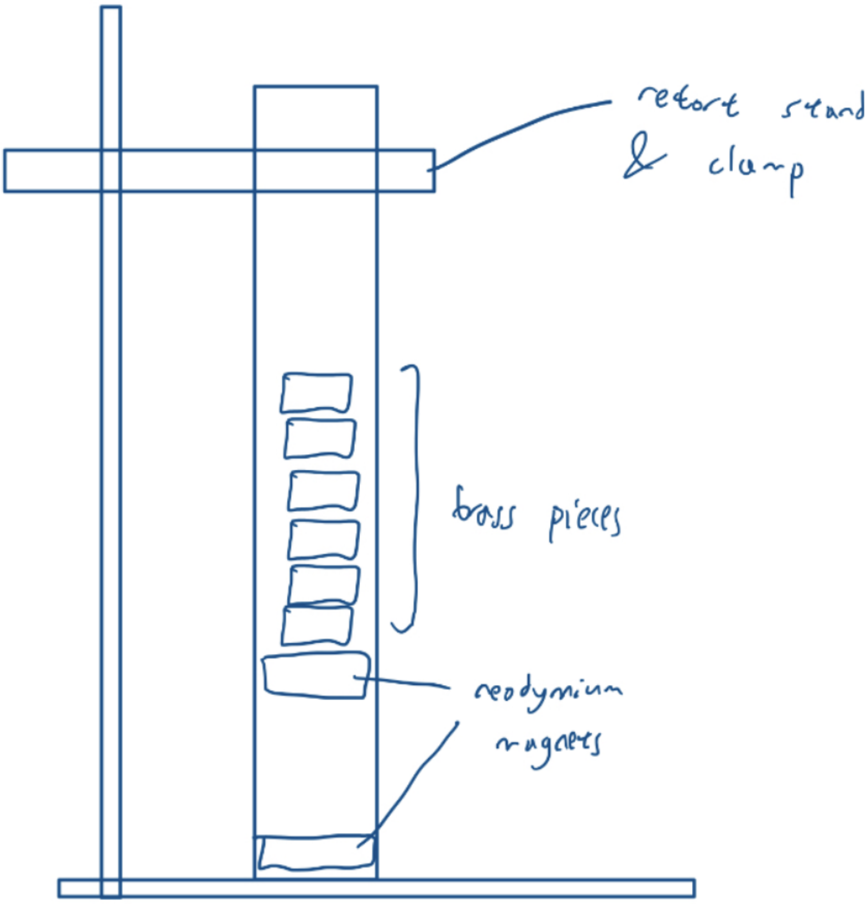
1.1 Apparatus Provided

1. 1 long, open tube with a cylindrical neodymium magnet attached at the bottom
2. 1 identical free neodymium magnet
3. 1 Jenga block with a neodymium magnet taped to it (meant for retrieval of the free neodymium magnet)
4. 1 long, non-magnetic cylindrical metal rod which fits into the tube
5. 15 cylindrical brass pieces of known density
6. 1 retort stand and clamp
7. 1 Vernier caliper (with precision up to 0.05 mm)
8. 1 tape measure
9. 1 stopwatch
10. Markers
11. Rubber bands

1.2 Part A

You are given the diagram of the experimental set-up (next page). Recreate the experimental set-up, and design and carry out an experiment to find:

1. the mass of a brass piece
2. the function $F(r)$ (the repulsive force between the two neodymium magnets when their like poles face each other)
3. the mass of the free neodymium magnet
4. the mass of the long metal bar



1.3 Part B1

Perform dimensional analysis to determine the dimensions of the quantity $u_{\mathrm{B}} = \frac{B^2}{2\mu_0}$. Based on its dimensions, comment on the significance of this quantity.

1.4 Part B2

Design and carry out an experiment to find the magnitude of the magnetic field B due to the neodymium magnet in terms of r, along the axial direction of the magnet. Plot a graph of B against r. In particular, indicate the value of B (in T) that you found at $r = 1.5\,\mathrm{cm}$.